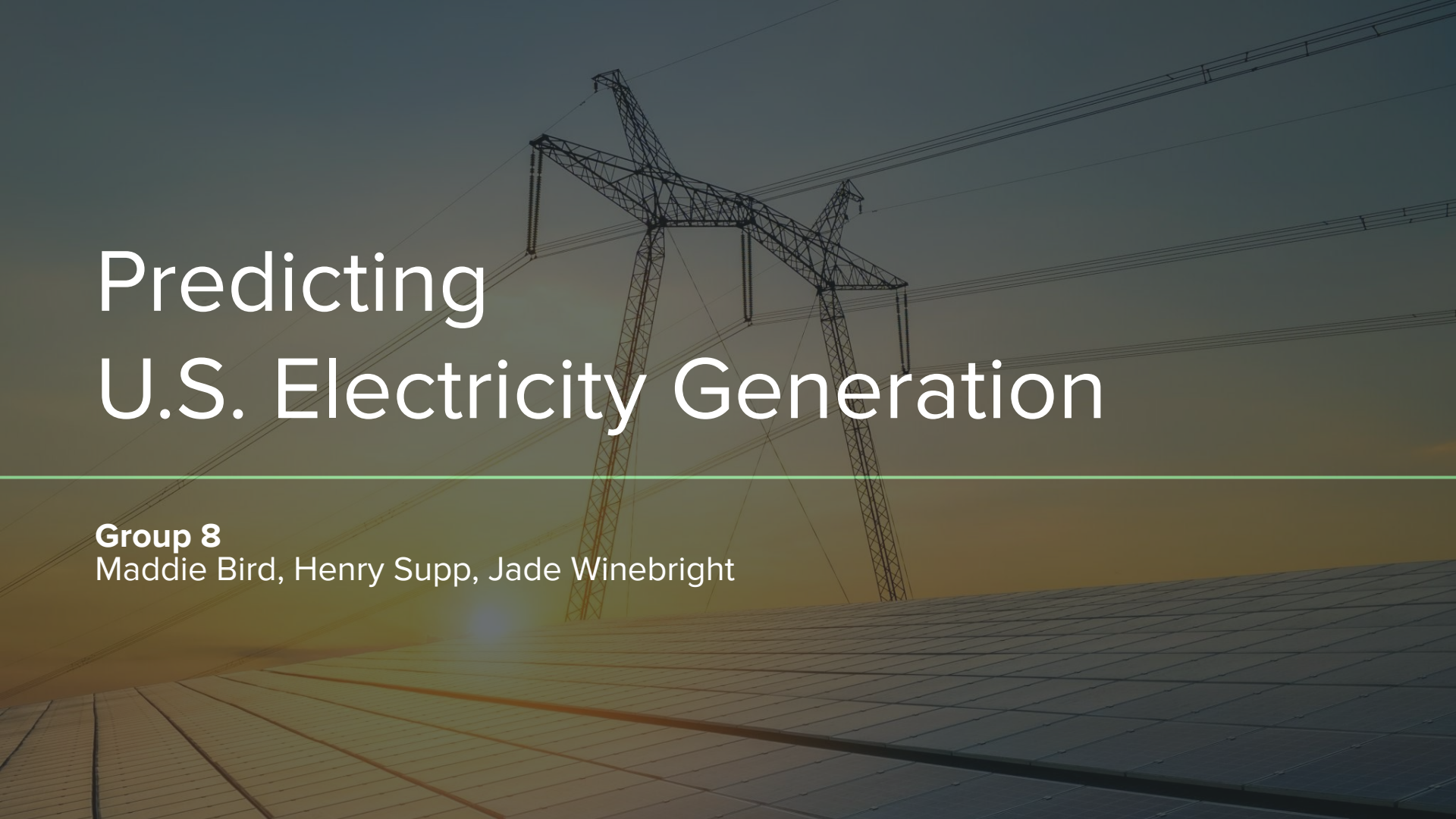




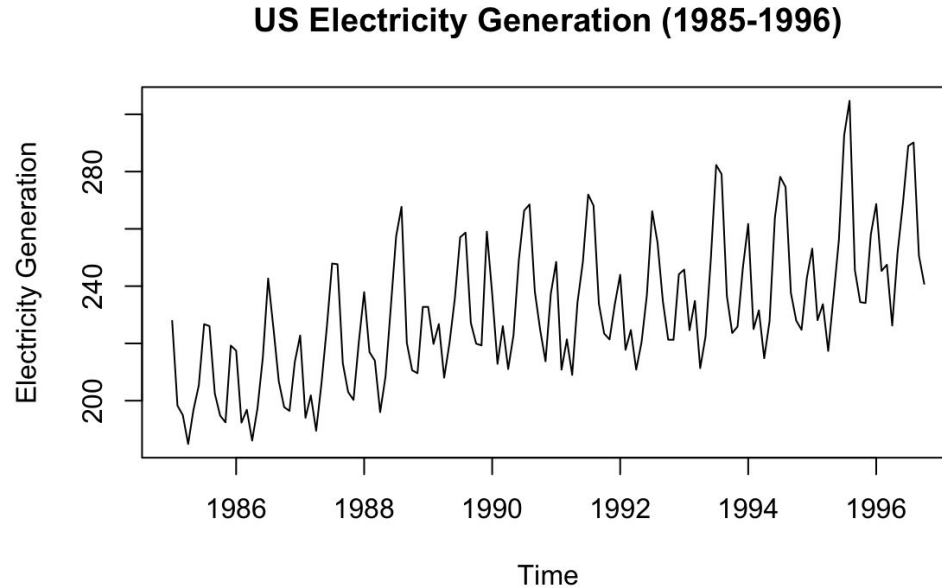
Predicting U.S. Electricity Generation

Group 8

Maddie Bird, Henry Supp, Jade Winebright

Data

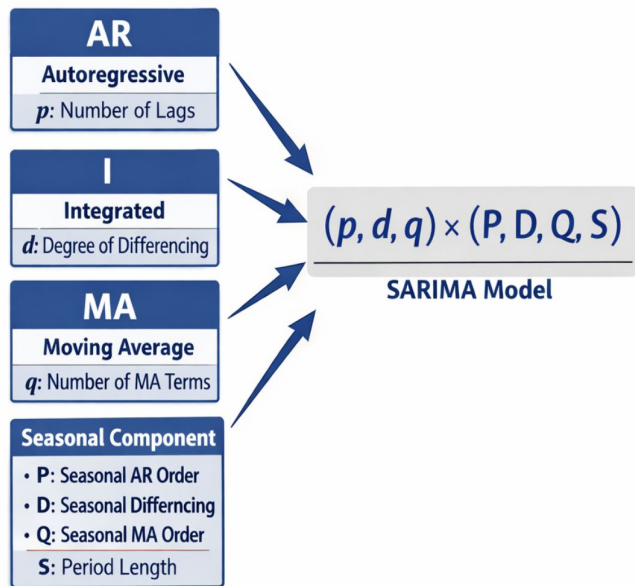
The monthly **total generation of electricity** by the U.S. electric industry (for Jan. 1985 – Oct. 1996)



Source: Makridakis,
Wheelwright and
Hyndman (1998)

Observations: 142

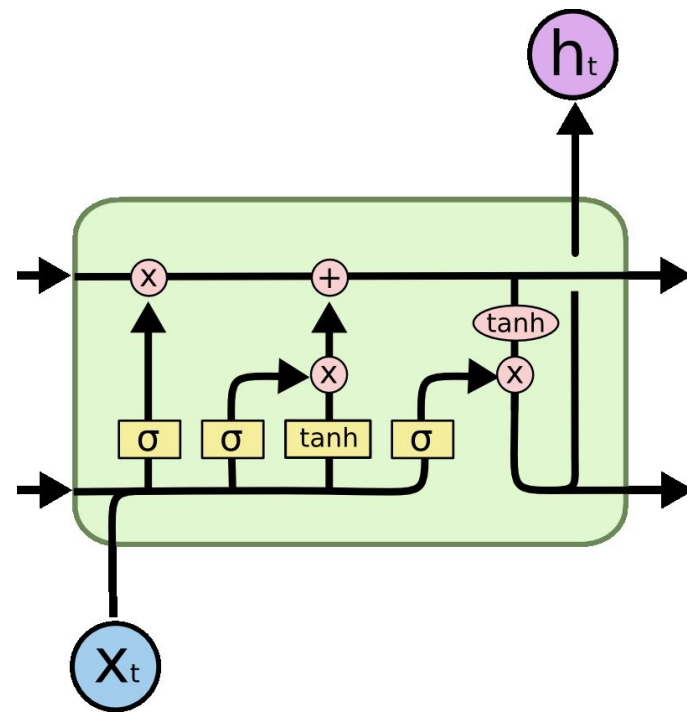
ARIMA Approach



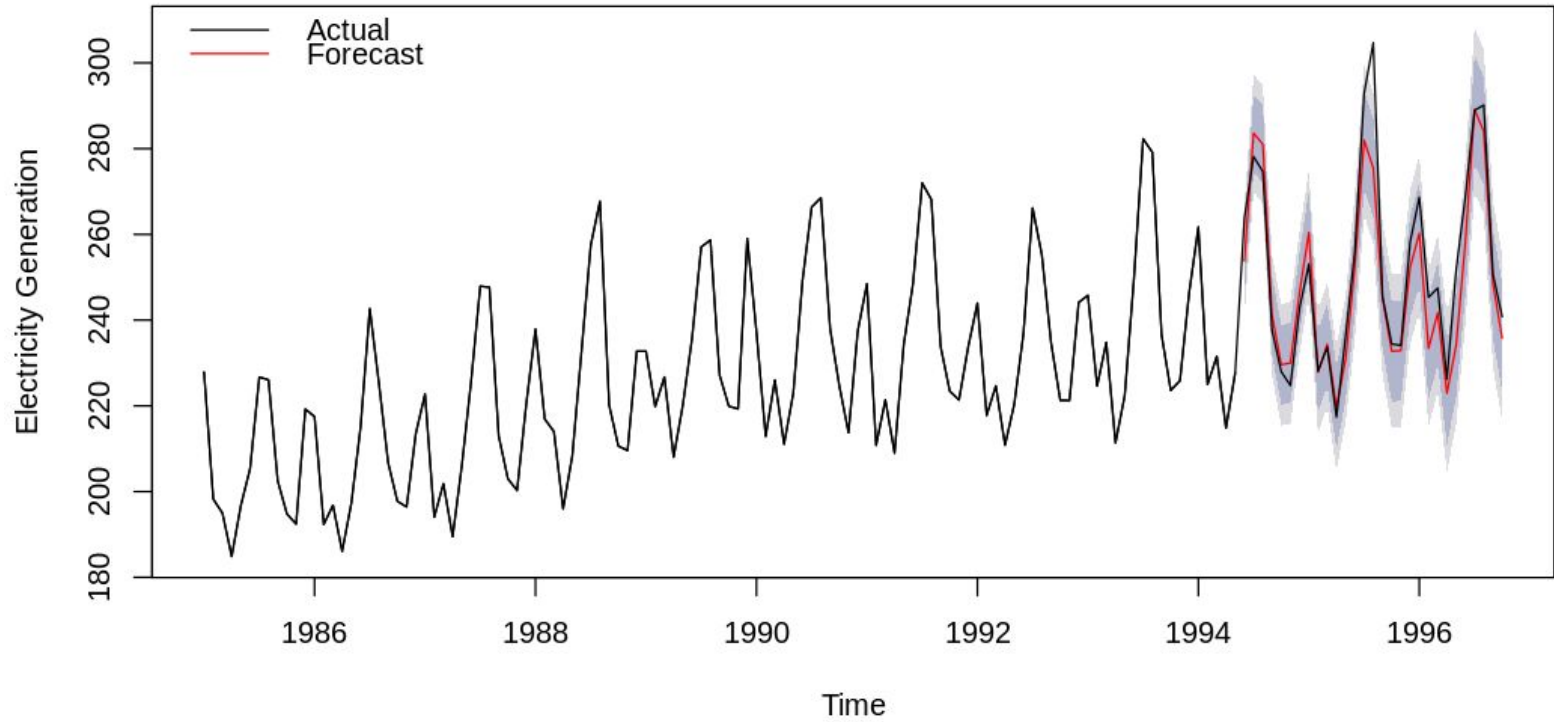
- Used Seasonal ARIMA (SARIMA) to capture trend and yearly (12-month) seasonality
- Split data into 80% training / 20% testing, preserving time order to avoid leakage
- Applied `auto.arima` to automatically select optimal $(p,d,q)(P,D,Q)_{12}$ parameters via AIC
- Model components include AR (past values), differencing (stationarity), MA (error correction), and seasonal terms

LSTM Approach

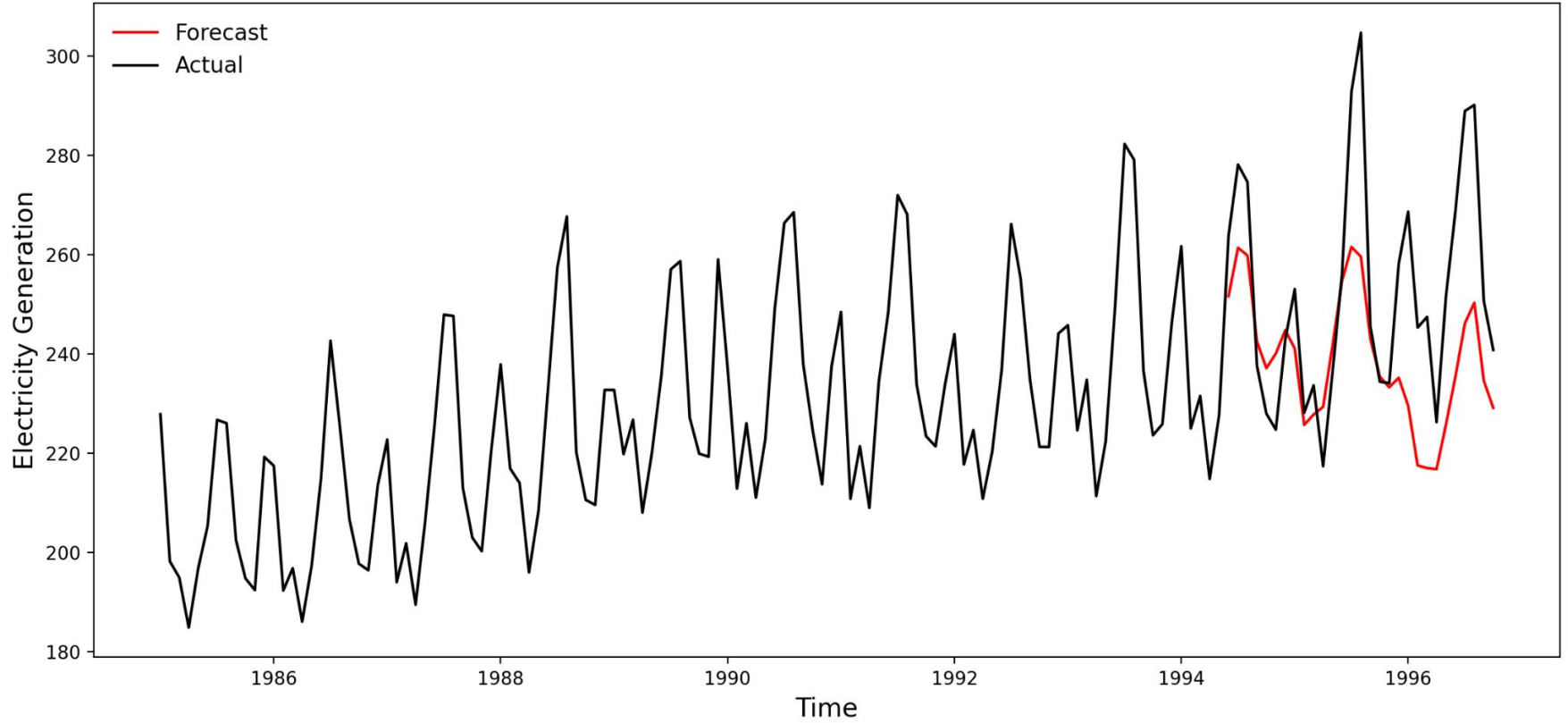
- Used a stacked LSTM using a 12-month sliding window to learn yearly seasonality.
- Split data into 80% training / 20% testing
- The network used
 - 32-unit LSTM layer, with L2 regularization and 0.2 dropout
 - 16-unit LSTM layer, with L2 regularization and 0.2 dropout
 - A final Dense layer produced one-step forecasts.
- Training used Adam, MSE loss, and EarlyStopping for stability.



ARIMA Forecast vs Actual



LSTM Forecast vs Actual



Metrics

- RMSE: Penalizes large errors → important for peak electricity demand
- MASE: Compares performance to a naïve baseline → useful for evaluating whether the model adds real predictive value

→ Combined: Provides a balanced & reliable evaluation

<u>Model</u>	RMSE	MASE	Best
ARIMA	8.484	0.844	✓
LSTM	27.946	2.724	

Overfitting Control

- Time-Based Split: Used 80% training / 20% testing split, preserving chronological order to prevent data leakage

ARIMA:

- Model order selected via AIC/BIC to avoid overfitting
- Differencing ensures stationarity and removes spurious patterns

LSTM:

- Applied dropout and L2 regularization to reduce model complexity
- Used early stopping based on validation loss
- Scaled data using training set only to prevent leakage

ARIMA vs LSTM

	ARIMA	LSTM
Generalization	More stable across time	More sensitive to data and training setup
Efficiency	Fast to train and forecast	More computationally intensive
Training Time	7.7 seconds	27.1 seconds
Tuning Effort/Sensitivity	Automated selection (auto.arima) → low tuning effort	Requires tuning architecture, hyperparameters → higher sensitivity
Interpretability & Deployment	Interpretable and easier to deploy	Black-box model, more complex to explain and implement

Conclusion

- ARIMA is better for small, seasonal datasets, LSTM requires more data to outperform.
- The dataset is extremely small (142 monthly points) → too limited for deep learning.
- U.S. electricity generation has strong, regular 12-month seasonality driven by heating and cooling demand. ARIMA directly models this structure using seasonal differencing and seasonal AR/MA terms.
- ARIMA: the most accurate, efficient, and interpretable model for this data.

